46. Node Metrics and Exporters (ops intro)

46. Node Metrics and Exporters (ops intro)

Phase: Phase 10 — Observability and troubleshooting
Estimated time: 1–2 days
Slug: `46-prometheus-node`

Learning objectives

- Expose node metrics conceptually
- Know RED/USE for hosts
- Alert on disk/inode/memory
- Avoid high-cardinality labels

Prerequisites

Complete prior numbered lessons. Prefer a real Linux VM/container for Linux-specific topics.

node_exporter + Prometheus/Grafana is a common Linux host monitoring stack.

Alert on:

- Disk >80–90%
- Inodes
- OOM events
- Systemd failed units
- Certificate expiry

Metrics without actionable alerts are wallpaper.

Common mistakes

- Paging on noisy low-value alerts
- No runbooks
- Cardinality explosions

Exercises

Work in `exercises/46-prometheus-node/`. Run:

```
./test.sh
```

- **A (Drill):** make `./test.sh` pass
- **B (Apply):** note one real command you ran on a Linux host related to this lesson
- **C (Challenge):** rewrite your solution from memory the next day

Do not open `solutions/46-prometheus-node/` until you have an honest failing attempt.

Self-check

1. Name three host alerts
2. USE method?
3. Why inode alerts?

Further reading

- https://prometheus.io/docs/guides/node-exporter/